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Mixture fraction imaging in turbulent non-premixed flames with two-photon LIF of krypton

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Abstract

The use of a noble gas as an inert tracer for mixing studies in combustion systems is investigated. Simultaneous two-photon laser-induced fluorescence (LIF) of krypton and Rayleigh scattering are used for imaging measurements of mixture fraction and temperature in turbulent non-premixed jet flames. The turbulent flames investigated in this study include a piloted CH_4/air flame (Sandia flame D) and a $CH_4/H_2/N_2$ flame (DLR-B flame). These flames are well-documented in the literature and enable an evaluation of krypton as a tracer in different fuel mixtures with varying degrees of differential diffusion. Krypton is excited from the ground state to the $5p[3/2]_2$ state using 215 nm laser radiation, and the fluorescence decay to the metastable state, $5s[3/2]_2$, is detected at 760 nm. Single-shot krypton LIF and Rayleigh scattering images are analyzed in an iterative routine to determine mixture fraction and temperature. Measurements of the temperature- and species-dependent quenching rates for Kr-LIF are incorporated into this routine. The resulting average radial profiles of mixture fraction and temperature for both flames agree well with previously published measurements. The use of a noble gas as a chemically inert tracer has potential applications for mixing studies in a broad range of combustion environments.
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1. Introduction

Mixing processes are central to turbulent combustion, and multi-dimensional measurements of mixing are needed to improve understanding of flow-flame interactions in turbulent non-premixed flames and stratified premixed flames. In non-premixed combustion, the state of mixing between the fuel and oxidizer streams is quantified by the mixture fraction, ξ, and the rate of molecular mixing is given by the scalar dissipation rate, defined as $\chi = 2D(\nabla\xi \cdot \nabla\xi)$, where D is the mixture-averaged mass diffusivity. Multi-dimensional measurements are required to determine the gradient of mixture fraction for calculating the scalar dissipation rate. Laser-based measurements of mixture fraction in flames are a challenge because of the variations in temperature and chemical composition. Single-point and line measurements of mixture fraction are feasible using simultaneous Raman/Rayleigh/CO laser-induced fluorescence (LIF) diagnostic techniques to measure all the major species concentrations and temperature [1,2]. However, this approach is not practical for

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multi-dimensional measurements. Two-dimensional mixture fraction imaging techniques include a two-scalar approach that combines Rayleigh scattering and fuel concentration measurements. Methods for obtaining fuel concentration have included LIF of fuel [3,4], Raman scattering from fuel [3–6], and difference Rayleigh scattering [7]. A three-scalar approach to mixture fraction imaging has also been developed using difference Rayleigh scattering and two-photon CO-LIF imaging [8,9].

In the present study, we consider an alternative method for measuring mixture fraction that uses a non-reactive tracer seeded into the fuel stream. A chemically inert tracer is, by definition, a conserved scalar and is only subject to passive mixing. The mixture fraction can be determined by combining measurements of the tracer concentration and temperature. This approach has previously been demonstrated using NO as the tracer and simultaneous measurements of NO LIF and Rayleigh scattering [10]. However, the use of NO as an inert tracer was limited to carbon monoxide/air flames because of NO chemistry in hydrocarbon flames. Here, we investigate the use of a noble gas as an inert tracer, which potentially has much broader applicability to mixing studies in combustion systems.

Of the noble gases, krypton and xenon are the primary contenders for mixture fraction imaging since they have electronic transitions that can be accessed using two-photon excitation in the UV. The lowest electronic transitions for neon and argon occur at approximately 18 eV and 13 eV, respectively, requiring tunable high intensity VUV radiation [11], which is impractical for measurements in optically thick combustion environments. Two-photon electronic transitions in krypton and xenon are accessible using commercially available lasers, which can produce 10–20 mJ/pulse at wavelengths of 200–300 nm. Several studies have documented the two-photon excitation of xenon at wavelengths of 200–220 nm [12–14]. However, krypton has the advantage of lower cost and atomic weight. The lower atomic weight is advantageous for reducing effects of differential diffusion between the tracer and other flame species.

Krypton LIF has been investigated by a number of groups. However, the use of krypton LIF for mixture fraction measurements has not been found in the literature. Quantitative krypton LIF measurements require knowledge of the temperature- and species-dependent collisional quenching rates, but the quenching rates for the $5p[3/2]_2$ state, which is used in the present study, are not available in the literature. In this manuscript, the states of krypton will be denoted using Racah $nl[K]_J$ notation. The $^2P_{3/2}$ and $^2P_{1/2}$ core states are denoted using unprimed and primed 1 values, respectively. Many of the previous Kr-LIF studies focused on metastable states of krypton, such as $5s[3/2]_1$ and $5s[3/2]_2$ [15–19]. Various groups have documented the quenching rates of metastable states with a number of collisional partners [16–18]. However, the quenching rates of the $5p[3/2]_2$ used in our studies could be significantly different from those of the metastable states. A few other studies have documented the use of two-photon LIF of krypton. Niemi et al. [20] used two-photon LIF excitation of the $5p'[3/2]_2$ state of krypton at 204.2 nm to calibrate measurements of atomic densities. This study included measurements of quenching rates for H_2, N_2, O_2, He, Ar, Kr, Xe, and CH_4. Again, since the excited state in that study differs from the one used in our study, these values should only be used as a guide. Results from studies using two-photon excitation of krypton for atomic density calibrations have compared well with other calibration methods [21,22]. Two-photon krypton excitation to $5p[3/2]_2$ has also been used in applications such as environmental monitoring [23].

Miller et al. [24] reported the use of multi-photon ionization (MPI) and detection of amplified stimulated emission (ASE) to document the transitions of krypton and xenon following two-photon excitation. Two possible excitation schemes of krypton include excitation to the $5p[5/2]_2$ and $5p[3/2]_2$ states at 216.6 nm and 214.7 nm, respectively. Through the use of MPI, it was shown that the $5p[3/2]_2$ produces more signal and therefore was chosen as the excitation wavelength in our study. The same study reports the branching ratio of the decay of the $5p[3/2]_2$ state to the metastable states $5s[3/2]_1$ (transition at 819.0 nm) and $5s[3/2]_2$ (transition at 760.2 nm) as 0.265 and 0.727, respectively. In our study, we detect fluorescence emission at 760 nm. The lifetime of the excited state $5p[3/2]_2$ was measured by Whitehead et al. [25] and Chang et al [26] to be 26.4 ± 0.5 ns and 25.4 ± 0.8 ns, respectively.

2. Experimental methods

2.1. Simultaneous krypton LIF and Rayleigh scattering imaging

The experiments were performed in the Advanced Imaging Laboratory at Sandia National Laboratories. A schematic drawing of the experimental configuration is shown in Fig. 1. It consisted of a combination of high-resolution Rayleigh scattering and krypton LIF imaging systems. The system used for imaging laser Rayleigh scattering has been thoroughly described in previous papers [27,28] and will only be briefly summarized here. Two frequency-doubled Nd:YAG lasers at 532 nm with energies of approximately 670 and 720 mJ/pulse each were combined into a single laser sheet. The resulting Rayleigh scattering was imaged onto an unintensified interline-transfer camera

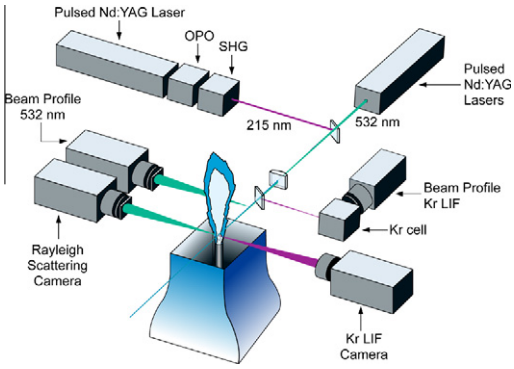

Fig. 1. Experimental setup for simultaneous two-photon krypton LIF and laser Rayleigh scattering imaging. OPO, optical parametric oscillator; SHG, second harmonic generation.

with an exposure time of 2 μs, bracketing the two Nd:YAG laser pulses. A second interline-transfer camera measured Rayleigh scattering from air in the co-flow adjacent to the probe volume to account for shot-to-shot variations in the Nd:YAG laser beam profiles.

The krypton LIF imaging system consisted of an optical parametric oscillator (OPO) pumped with a frequency-tripled Nd:YAG laser. The output of the OPO was frequency doubled to produce 11–14 mJ/pulse of UV light, which was tuned to probe the Kr($^1S_0$) to Kr(5p[3/2]$_2$) transition at 214.7 nm. The beams from the OPO and the two frequency-doubled Nd:YAG lasers were combined by passing the 532-nm beams just above the 215 nm turning mirror at a slight downward angle, such that the beams overlapped in the test section. The combined beams were focused into a sheet using a single 500-mm focal length cylindrical lens. The resulting krypton LIF emission at 760 nm was imaged onto a Gen III intensified CCD camera using back-to-back medium and 35-mm format commercial camera lenses and a 760-nm narrow-band filter ($\Delta\lambda = 10$ nm). The intensified camera was gated for 700 ns bracketing the OPO laser pulse. A quartz plate diverted 8–10% of the UV laser energy into a beam profiling cell that contained 1% Kr in He at 294 K. The resulting Kr-LIF signal was imaged onto an unintensified camera to monitor shot-to-shot variations in the OPO beam profile. The timing of the experiment was as follows: the laser pulses for excitation of Rayleigh scattering were separated by 700 ns, followed by the UV laser 1 μs later.

The non-premixed turbulent jet flames considered are the DLR-B flame [29], and the Sandia flame D [1], which are target flames for turbulent combustion modeling in the TNF Workshop [30]. These flames are well-documented in previous studies and single-point Raman/Rayleigh scattering measurements of species concentrations, temperature, and mixture fraction are available to validate the current diagnostic technique. In addition, the different composition of the fuel mixtures of these flames provides a test of the effects of differential diffusion between krypton and the fuel. For the DLR flame, the fuel was a mixture of methane, hydrogen, and nitrogen mixed in mole fractions of 22%, 33%, and 45%, respectively, giving a stoichiometric mixture fraction of 0.167. For this fuel mixture, the Rayleigh scattering cross section was constant to within ±3% throughout the flame, enabling direct determination of the gas number density from the Rayleigh scattering signal [31]. The mixture exited the nozzle ($d = 8.0$ mm) at a Reynolds number 22,800 into a filtered air co-flow ($u = 0.3$ m/s). The fuel in the piloted flame consisted of 25% CH_4 and 75% air (by volume), resulting in a stoichiometric mixture fraction of 0.351. The premixed pilot flame was a mixture of C_2H_2, H_2, dry air, CO_2, and N_2. The jet nozzle diameter was 7.2 mm, and the jet flow exited the nozzle at a Reynolds number of 22,400 into an air co-flow ($u = 0.9$ m/s). For both flames, krypton was seeded into the fuel stream at a mole fraction of 1%. This seeding level avoided disturbance of the flame and minimized interferences from stimulated emission.

The resolution of the mixture fraction measurements was limited by the intensified camera. The projected pixel size for the Rayleigh scattering images was 31 × 31 μm with 2 × 2 on-chip binning and 52 × 52 μm for the Kr-LIF images. The imaged area of the Rayleigh scattering measurement was significantly larger than that of the Kr-LIF measurement as a result of the small UV beam height that was required for excitation of two-photon LIF. The overlapping imaged area for the combined Rayleigh scattering and krypton LIF images was approximately 21 × 5.5 mm. Registration of the Rayleigh scattering and Kr-LIF images was accomplished using a target image that was recorded on both cameras. The data reduction for the krypton LIF and Rayleigh scattering images included corrections for shot-to-shot variations in the respective beam profiles and the optical throughput of the imaging systems.

2.2. Collisional quenching, absorption, and stimulated emission

The effects of collisional quenching, absorption, and stimulated emission (SE) on the krypton LIF measurements were evaluated. Variations in temperature- and species-dependent collisional quenching rates within the flame bias the krypton LIF signal and must be accounted for in the calculation of mixture fraction. Quenching rates were measured at temperatures up to 850 K by combining two different experimental approaches, and the results were extrapolated to flame

temperatures. Absolute quenching rates were measured in a low-pressure cell at 294 K, and relative quenching rates were measured at atmospheric pressure in a heated flow from 294 to 850 K. The relative quenching rates were then calibrated by matching the values at 294 K to those obtained from the absolute measurements. The collisional partners considered were Kr, N_2, H_2, CO_2, O_2, H_2O, and CH_4.

The absolute quenching rates at 294 K were determined from measurements of fluorescence decay times in a cell that was filled with 200 mTorr of krypton and up to 95 Torr of the quencher gas. Two-photon LIF of krypton was excited using the frequency-tripled amplified output of a distributed feedback dye laser (DFDL) operating at a repetition rate of 20 Hz with a pulse duration of approximately 80 ps, a linewidth of $0.6\,\mathrm{cm}^{-1}$, and tuned to 214.7 nm. The DFDL and amplifiers were pumped by the second harmonic of a regeneratively amplified Nd:YAG laser, which was seeded by a mode-locked laser. Details of the DFDL system are described in Refs. [32,33]. The resulting Kr-LIF emission was collected at 90 degrees to the laser beam and focused onto a microchannel plate photomultiplier tube (MCP-PMT) using a pair of achromatic lenses and a 760 nm bandpass filter ($\Delta\lambda = 10$ nm) to block out-of-band interferences.

The decay rate of the excited state of krypton, $5p[3/2]_2$, is determined by the quenching rate of each collisional partner in the gas mixture and the radiative decay of the excited state,

$$\gamma = A + \sum_i q_i X_i, \tag{1}$$

where A is the spontaneous emission rate, q_i and X_i are the temperature-dependent quenching rate and mole fraction of species i, respectively. For each quenching gas, the LIF signal decays were recorded in a sequence of increasing quenching gas pressure. At each pressure, the average of 12,800 single-shot LIF decay traces was fit to a single-exponential decay. The self-quenching rate, q_{kr}, was determined from measurements of the LIF decay rate as a function of krypton pressure in the absence of additional quenchers. The spontaneous emission rate was extracted from these measurements by extrapolating to zero krypton pressure. The quenching rate of each gas was calculated using a linear curve fit of the measured decay rates as a function of quencher pressure at constant krypton pressure. Extra care was taken to avoid stimulated emission since it can adversely affect the fluorescence decay rate measurements. To this end, an unfocused laser beam was used, and the maximum laser pulse energies and Kr concentrations were 50 μJ and 200 mTorr, respectively.

Relative quenching rates were determined from measurements of time-integrated Kr-LIF signals in heated atmospheric pressure laminar flows containing 0.2–0.6% (by volume) krypton and different concentrations of collisional partners with helium or nitrogen as a buffer gas. Helium is a significantly weaker quencher than the other gases considered here. The experimental apparatus for the quenching measurements consisted of a 25 mm diameter fused silica tube fitted with a SiC heating coil. The LIF excitation and detection configuration was identical to that of the Kr-LIF imaging setup described in Section 2.1. The laser sheet was focused into the heated flow at the exit of the fused silica tube. The Kr-LIF signal was measured with the intensified camera and was corrected for variations in laser energy via the beam profiling camera. The Stern–Volmer relation for each quenching gas was determined from a sequence of LIF measurements with different quencher concentrations.

The krypton quenching rate by species i in a bath of helium can be determined relative to the quenching rate of pure helium using the Stern–Volmer relation, where the ratio of fluorescence decay lifetimes of krypton, τ, is proportional to the ratios of fluorescence signals, S:

$$\frac{\tau_{\mathrm{He}}}{\tau_i} \propto \frac{S_{\mathrm{He}}}{S_i} = 1 + \alpha\left(\frac{q_i}{q_{\mathrm{He}}} - 1\right)X_i. \tag{2}$$

S_{He} is the Kr-LIF signal in pure helium, S_i is the LIF signal in helium with quencher i added, $\alpha = \left(\frac{A}{q_{\mathrm{He}}} + \frac{q_{\mathrm{Kr}}}{q_{\mathrm{He}}}X_{\mathrm{Kr}} + X_{\mathrm{He}}\right)^{-1}$ is a function of the krypton and helium mole fractions in the flow without any additional quencher. The ratio of the quenching rates, $\frac{q_i}{q_{\mathrm{He}}}$, can be calculated from the measured slope of $\frac{S_{\mathrm{He}}}{S_i}$ plotted as a function of the quencher mole fraction, X_i, at a particular temperature. The ratio of Kr-LIF signals for helium and nitrogen at atmospheric pressure and room temperature indicate that the helium quenching rate is approximately a factor of 7–8 less than that of N_2. The temperature dependence of the quenching rates was determined by repeating these measurements at four different temperatures from 294 to 850 K, using the absolute quenching rate measurements to calibrate the relative quenching rate measurements at 294 K. The resulting quenching rates were fit using a power-law temperature dependence.

The interpretation of the two-photon LIF signals can be complicated by laser-generated interferences. We verified that laser-generated interferences were negligible in these flames by tuning the OPO laser off resonance. The effects of stimulated emission and absorption were investigated to determine the maximum useable krypton seeding concentration. Stimulated emission was directly measured using a beam profiling camera to detect forward scattered 760-nm emission after the OPO beam passed through the flame. A dichroic mirror and a 695-nm long-pass filter rejected the 215-nm light from the laser beam.

The dichroic mirror diverted the laser beam into a laser energy meter. The laser energy was measured with and without krypton seeding to verify that absorption of the laser beam was negligible for the flames considered here. The intensified CCD camera described above was used to measure the Kr-LIF signal along the path of the beam as it propagated through the flame to determine the effect of SE on the LIF measurements.

3. Results and discussion

3.1. Mixture fraction calculation

The mixture fraction is determined from the krypton mole fraction on a pixel-by-pixel basis using the Kr-LIF signal, S_{Kr}, and the Rayleigh scattering signal, S_{Ray}, in an iterative algorithm. First, the mixture fraction is related to the mole fraction of gases originating from the fuel stream, X_{fuel}, by

$$\xi = X_{fuel} * \frac{W_{fuel}}{W_{mix}(\xi)}, \tag{3}$$

where W_{fuel} is the molecular weight of the fuel stream, and $W_{mix}(\xi)$ is the local mixture molecular weight. The mole fraction of gases from the fuel stream is

$$X_{fuel} = \frac{S_{Kr}}{S_{Kr,fuel}} \frac{\gamma(\xi, T)}{\gamma_{fuel}} \frac{T}{T_{fuel}}, \tag{4}$$

where $S_{Kr,fuel}$, γ_{fuel}, and T_{fuel} are the krypton LIF signal, fluorescence decay rate, and temperature of the fuel stream, respectively. The fluorescence decay rate in the flame, $\gamma(\xi, T)$, is defined by Eq. (1) and is a function of the local mixture fraction and temperature. In the flames considered here, the decay rates are predominantly determined by the quenching rates of the collisional partners, which are two orders of magnitude greater than the spontaneous emission rate. The temperature is determined from the Rayleigh scattering signal as follows:

$$T = \frac{S_{Ray,ref}}{S_{Ray}} \frac{\sigma_{Ray}(\xi)}{\sigma_{Ray,ref}} T_{ref} \tag{5}$$

where T_{ref}, $\sigma_{Ray,ref}$, and $S_{Ray,ref}$ are the temperature and Rayleigh scattering cross section and signal in a reference gas, such as air.

An iterative procedure is required for determining mixture fraction because W_{mix} and σ_{Ray} are functions of mixture fraction and γ is a function of both mixture fraction and temperature. The functional dependence of these parameters was determined using polynomial fits to data from previous multiscalar point measurements of these flames coupled with the Kr-LIF quenching rate measurements in the present study. For the iterative procedure, an initial estimate of the mixture fraction and temperature were made and initial values of W_{mix}, γ, and σ_{Ray} were determined from the curve fits. Note that in the DLR-B flame, σ_{Ray} was considered a constant. In subsequent iterations, the mixture fraction and temperature were updated using Eqs. (3)–(5). These updated values were then used to recalculate W_{mix}, γ, and σ_{Ray}. This process was repeated until subsequent calculations of mixture fraction differed by less than 0.01.

3.2. Turbulent jet flames

Mixture fraction and temperature measurements were performed using simultaneous imaging of Kr-LIF and Rayleigh scattering in the two turbulent flames of interest. In each flame, 600 image pairs were acquired. For the DLR-B flame, measurements were performed at a downstream position of $x/d = 10$ spanning radial positions $r/d = -0.8$ to 1.8. For the Sandia flame D, two adjacent measurement locations centered at $x/d = 30$ spanned radial positions of $r/d = -1.0$ to 5.0. The signal-to-noise ratio of the Kr-LIF signal in the cold fuel was on the order of 40 and approximately 10–12 in the stoichiometric region of a laminar flame. The total quenching rate for lean and rich conditions was calculated as a function of temperature using the measured quenching rates of major species and previous multiscalar point measurements [1,29]. The resulting scatter plots of normalized LIF decay rates, γ/γ_{fuel} for lean and rich conditions were fit using separate polynomial functions for the Sandia flame D and DLR-B flame, as shown in Fig. 2. These fits were used to iteratively calculate mixture fraction as explained above, requiring on average less than five iterations per pixel to converge.

Sample single-shot results from the Sandia flame D are shown in Fig. 3. These Kr-LIF and Rayleigh scattering signals are corrected for background and variations in the laser beam profile. Corresponding mixture fraction and temperature measurements are shown in the bottom row. To verify the accuracy of the mixture fraction and temperature measurements, we compare mean radial profiles of mixture fraction and temperature from the present study with previous multiscalar point measurements obtained via Raman and Rayleigh scattering [1,29]. These comparisons are shown in Figs. 4 and 5 for Sandia flame D and the DLR-B flame, respectively.

The agreement between the current diagnostic approach and previous point measurements is quite good for Sandia flame D. The imaging measurements slightly overshoot the centerline peak mixture fraction of approximately 0.66, and undershoot the centerline temperature of 1340 K. However, the agreement is well within the uncertainties that result from errors in the quenching rates. The locations of the peak values and the

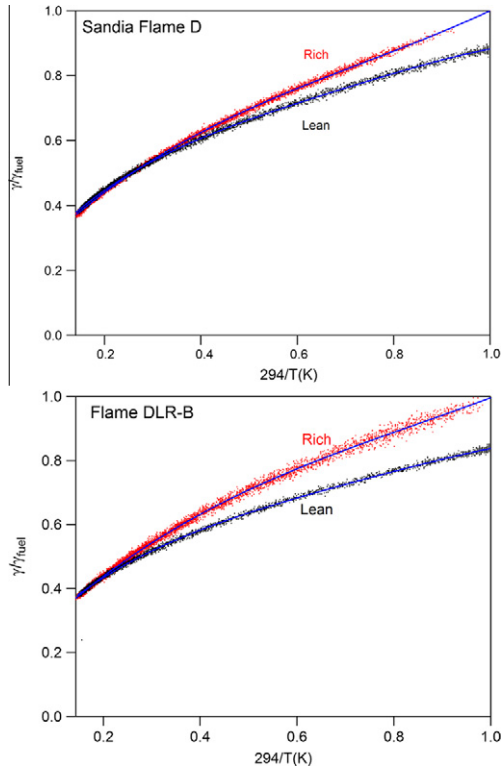

Fig. 2. Relative fluorescence decay rates plotted as a function of temperature for rich and lean conditions with polynomial fits for Sandia flame D (top) and DLR-B flame (bottom).

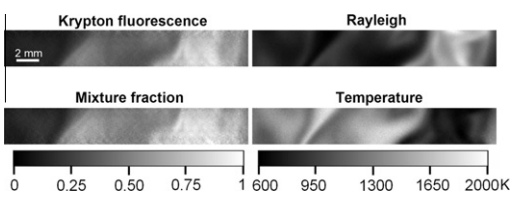

Fig. 3. Single-shot krypton LIF and Rayleigh scattering images from Sandia flame D (top row). Resulting mixture fraction and temperature measurements (bottom row).

radial variation match well from lean to rich conditions, and the peak temperature of approximately 1700 K shows excellent agreement. For flame DLR-B, there is also good overall agreement. The centerline values for the mixture fraction and temperature agree with the single-point measurements, and the rise of temperature through the reaction zone follows the point measurements. The slightly narrower width of the mixture fraction profile in the imaging measurements may result from differential diffusion between krypton and the fuel. Differential diffusion is enhanced in the DLR-B flame as a result of the H_2 in the fuel stream. The

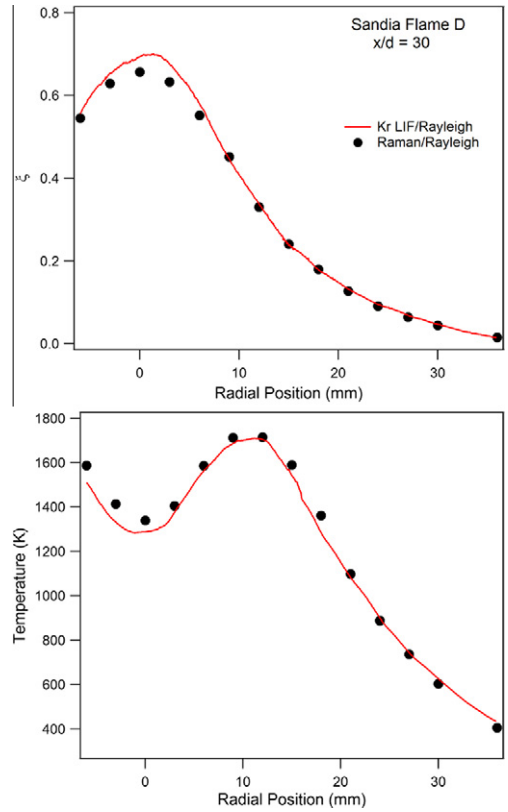

Fig. 4. Comparisons of mixture fraction and temperature profiles measured in this study and previous single-point measurements in Sandia flame D at $x/d = 30$ [1].

uncertainties in quenching rates at flame temperatures may also contribute to these discrepancies.

3.3. Effects of stimulated emission and collisional quenching

The effects of stimulated emission on the Kr-LIF signals were investigated in the DLR-B flame and the Sandia flame D, as well as in corresponding laminar jet flames with fuel mixtures identical to those of the turbulent flames. The laminar flames were stabilized on the same nozzles as the turbulent flames, and measurements were performed 3.0 cm downstream of the jet exit. The SE measurements were performed using a laser energy of 11 mJ/pulse and a laser sheet height of 5.5 mm. Direct detection of forward scattered stimulated emission was combined with measurements of the mean radial profiles of the Kr-LIF signal in the flame, as described in Section 2.2. Previous studies of two-photon LIF of atomic oxygen and hydrogen in flames showed that stimulated emission can distort LIF signal profiles across a flame [34,35].

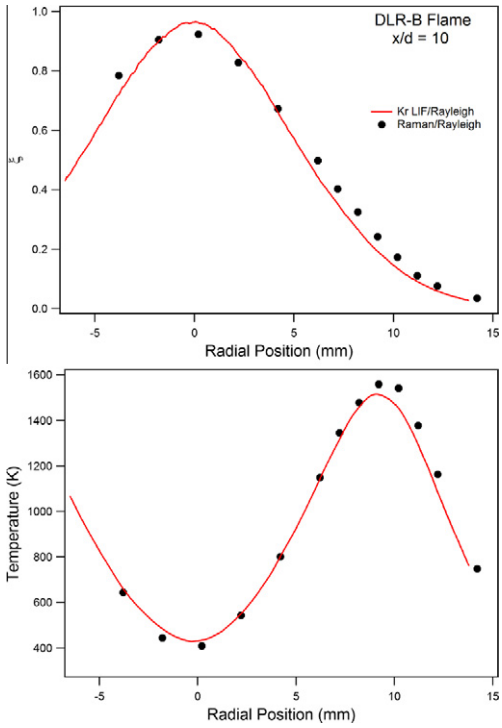

Fig. 5. Comparisons of mixture fraction and temperature profiles measured in this study and previous single point measurements in the DLR-B flame at $x/d = 10$ [29].

Table 1
Power-law temperature dependence of quenching rates obtained from relative quenching rate measurements in atmospheric pressure heated flows combined with calibrations at 294 K using low-pressure cell measurements.

Species	$q(T) = q_0(T/294)^n$	
	$q0$ (MHz/Torr)	n
N_2	6.5	−0.26
H_2	18	−0.53
CO_2	24	−0.78
O_2	25	−0.64
CH_4	17	−0.71
H_2O	47	−0.96
Kr	1.0	–

In the laminar flames, a sequence of measurements were performed using krypton mole fractions from 0.001 to 0.20. The threshold for SE occurred at a mole fraction of 0.005. However, undistorted radial profiles of Kr-LIF signals were measured for krypton mole fractions up to 0.04. For mole fractions greater than 0.04, SE distorted the profiles of the LIF signal in the flame. In the turbulent DLR-B flame, the SE threshold occurred at a similar Kr mole fraction to that of the laminar flames. Although SE was detected at the mole fractions used in this study (0.01), distortion of the radial profiles of the LIF signal in the jet flames was negligible. At higher mole fractions, SE resulted in a narrowing of the radial profiles of the LIF signal in the flames. The dependence of SE on krypton concentration, laser intensity, and laser path length through the flame is highly nonlinear. As a result, the SE is dependent on the krypton seeding concentration and flame geometry. Additional studies demonstrated that high seeding concentrations of krypton enable the use of an unintensified camera for qualitative visualization of mixing in flames using two-photon Kr-LIF imaging. However, significant SE distorts the measured jet profiles and precludes quantitative measurements.

The inclusion of collisional quenching rate variations in the mixture fraction calculations is important for attaining accurate results. Table 1 summarizes the measured power-law temperature dependences of the quenching rates. The low-pressure, ambient temperature fluorescence decay rate measurements were used to determine q_0, and the Stern–Volmer relations from the heated, atmospheric pressure flows were used to determine the power-law exponent. Aside from krypton self-quenching, N_2 is the least efficient quencher listed. The quenching rates of all species decreased with increasing temperature. The quenching rates of CH_4, H_2, CO_2, and O_2 are grouped relatively closely. The quenching rates of these species are approximately 30–50 times that of He at 294 K. Water vapor is the most efficient quencher and has the strongest temperature dependence. Accurate measurements of q_{H_2O} are challenging and have the largest uncertainty. For the low-pressure cell measurements, H_2O was introduced as a quencher by pulling the vapor from a second, independently pumped, vacuum cell containing distilled water. The temperature dependence of q_{H_2O} was determined by extrapolating these low-pressure cell experiments at 294 K to flame temperatures. The extrapolation is based on measurements of the ratio of Kr-LIF signals in an atmospheric 294 K air flow and the combustion products of a premixed H_2/air flame stabilized on a Hencken burner. Current efforts are focused towards refining the quenching rate measurements and obtaining absolute measurements at elevated temperatures.

4. Conclusions

The use of a noble gas as a chemically inert fluorescent tracer for combustion systems was demonstrated. Simultaneous two-photon laser-induced fluorescence of krypton and Rayleigh scattering were used for single-shot mixture fraction and temperature measurements in turbulent non-premixed jet flames: Sandia flame D and the

DLR-B flame. For the Sandia flame D, the agreement of mixture fraction and temperature profiles with previously published Raman/Rayleigh point measurements was excellent. For the DLR-B flame, the temperature profile agreed well, but there was a slight systematic discrepancy in the DLR-B mixture fraction profile width, which is indicative of differential diffusion. The presence of hydrogen in the DLR-B fuel mixture leads to increased differential diffusion.

The effects of collisional quenching by N_2, H_2, CO_2, O_2, H_2O, and CH_4 were investigated. Measurements of relative quenching rates from 294 to 850 K were coupled with absolute quenching rate measurements at 294 K. N_2 has the lowest quenching rate, and H_2O has the highest quenching rate. The quenching rates of all species decrease with temperature, and the results indicate that H_2O has the strongest temperature dependence. Additional measurements are needed to confirm the temperature dependence of the quenching rates.

Stimulated emission limited the useable krypton seeding mole fraction for quantitative measurements to a few percent. At higher mole fractions, stimulated emission distorted the measured radial profiles in the flames. The onset of stimulated emission has a nonlinear dependence on the krypton concentration and the path length of the laser through the flame and is therefore dependent on the flame geometry.

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